Zain Dana Harper

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AI Systems Engineering | Evaluation | Python Developer Tools

Summary

Self-directed software builder transitioning from arboriculture into AI evaluation and Python tooling, with maintainer-merged fixes for evaluation crashes, configuration data loss, and API errors.

Technical Skills

Python, Rust, C++, Dart/Flutter, SQL, GitHub Actions, pytest, Linux, Windows; AI evaluation, MCP, APIs, deterministic testing, typed capabilities, approval gates, offline verification.

Self-Directed Projects

About 13 years self-directed | Career transition began June 2026

- Built and shipped Flywheel 1.9.0 on PyPI, a Python/Flutter workstation with Rowan orchestration, model routing, tools, and a Windows installer; its behavior-check caught 20 of 20 planted changes with 0 of 50 false alarms.
- Built 324 exhaustive terminal-state fixtures and 337 synthetic component cases with 5,058 frozen expectations, distinguishing failed outputs from valid results within defined schemas.
- Built Articulate 0.14.0 writing analysis, Accountable Surface approval interfaces, BuildLang 1.8.0 capability checks, and Phantom audit/restore workflows for inspectable changes.
- Executed a pinned METR count_odds check with three deterministic controls, retaining logs and score provenance through the Inspect-to-Flywheel handoff.

Freelance Technical Writer | Independent projects | Began 2017

- Wrote implementation guides, API documentation, and release notes for developers and customers.

Employment

Arboriculture and Field Operations | Legendary Tree

Seattle area | April 25, 2015 to June 2, 2026

- Turned site assessments into estimates, proposals, and workable job scopes; coordinated clients, crews, and vendors so teams understood access constraints, timing, and hazards.

Technical Networking Support | Stream/Convergys | 2014 to 2015

Xbox/Microsoft product contract | Wilsonville, Oregon

- Resolved Xbox connectivity problems across TCP/IP, DNS, NAT, firewalls, and routers; documented troubleshooting steps for consistent support handoffs.

Selected Open-Source Contributions: 6 Merged

- DeepEval #2822: fixed crashes on unhashable tool outputs and nested parameters; added regression tests to protect evaluation reliability. Merged.
- TOMLKit #549: preserved sibling keys during dotted-key replacement, preventing silent configuration data loss on round trips. Merged.
- Datasette #2815: corrected malformed composite-key requests from HTTP 500 or incorrect HTTP 200 to HTTP 400, with regression coverage. Merged.

Credentials, Training, and Education

Qualys Certified Specialist: Policy Audit, CAR, VMDR, KCS, TotalCloud, EDR. Microsoft Applied Skills: AI, identity, storage, and Azure DevOps security (95% assessment). Google badges/coursework: language-model coding, multi-agent systems, BigQuery prediction and modeling, MLOps evaluation. Coursework: security, AI governance, and developer tools. Full record: harperz9.github.io/cv.html.

High School Diploma | Wilsonville High School, Oregon | 2013